

C-DOT control science

Phase 2.1 → Phase 3.2 · v6

Post-audit safety and evidence corrections

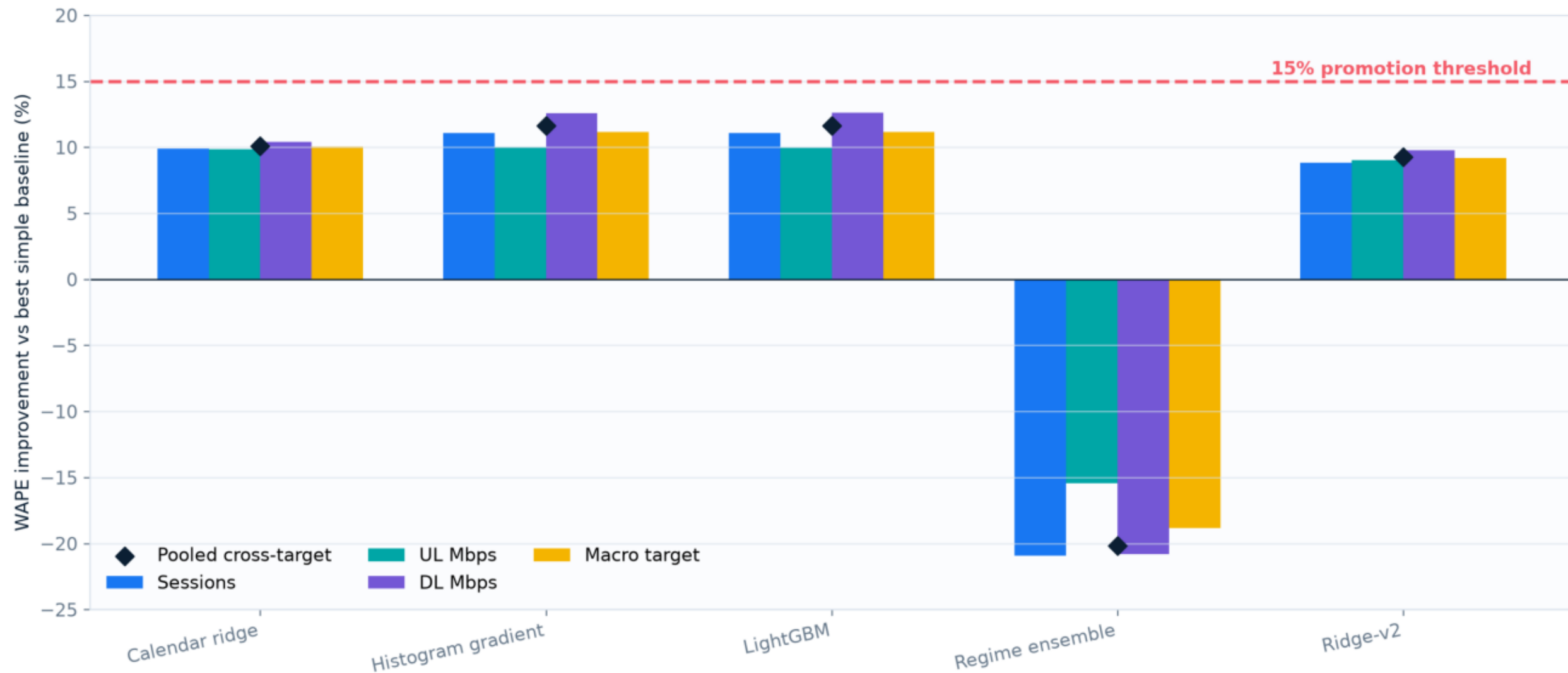
Exact evidence scale

516 candidate pairs + 72 sensitivity pairs = 588 total

Outcome: Static production · candidates guarded shadow/replay only

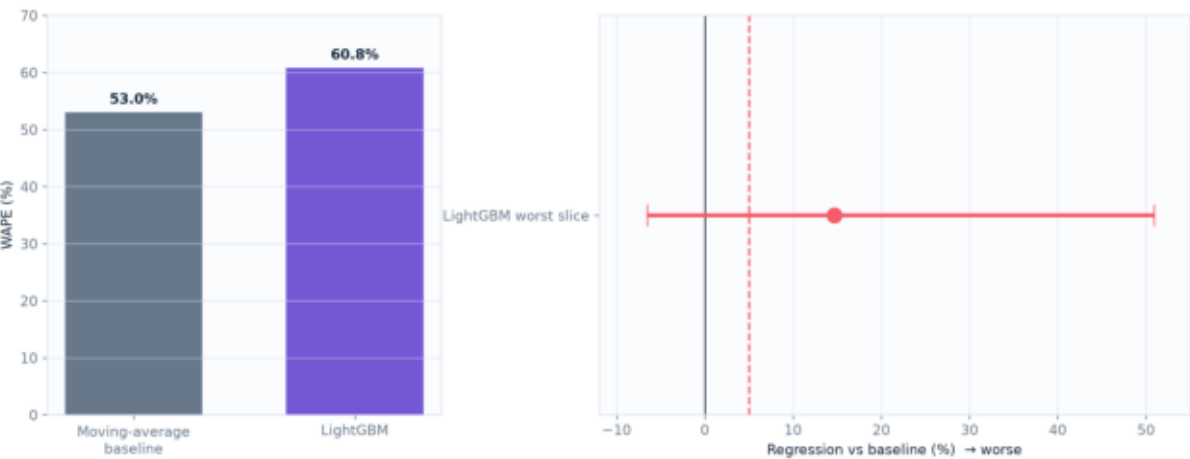
Forecast gains are real—but below the promotion bar

Target-separated WAPE prevents unlike units from being hidden inside the pooled headline



The worst LightGBM slice is visible—and statistically fragile

Detected-surge DL load at 30 minutes: n=9 observations across 8 groups

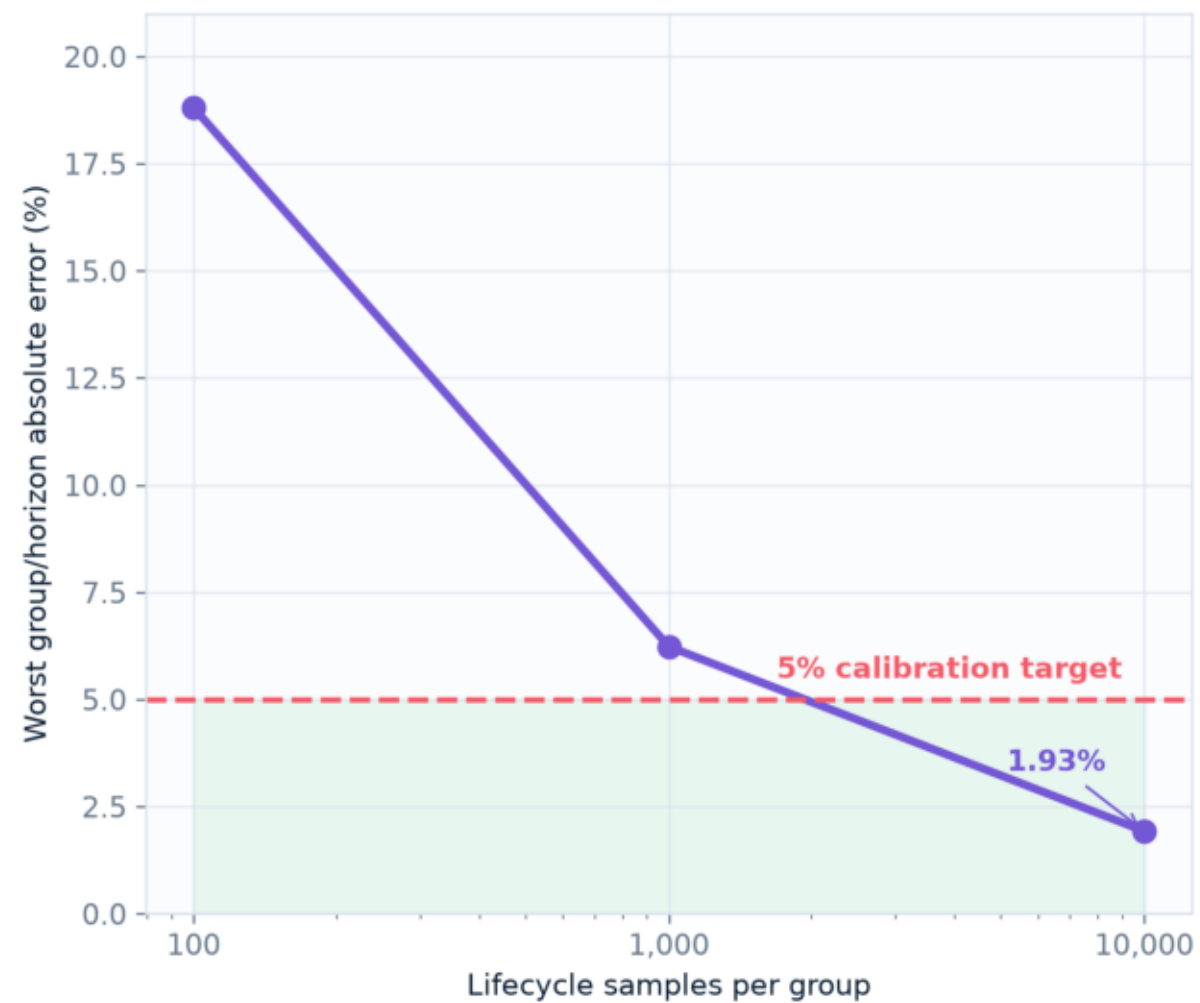
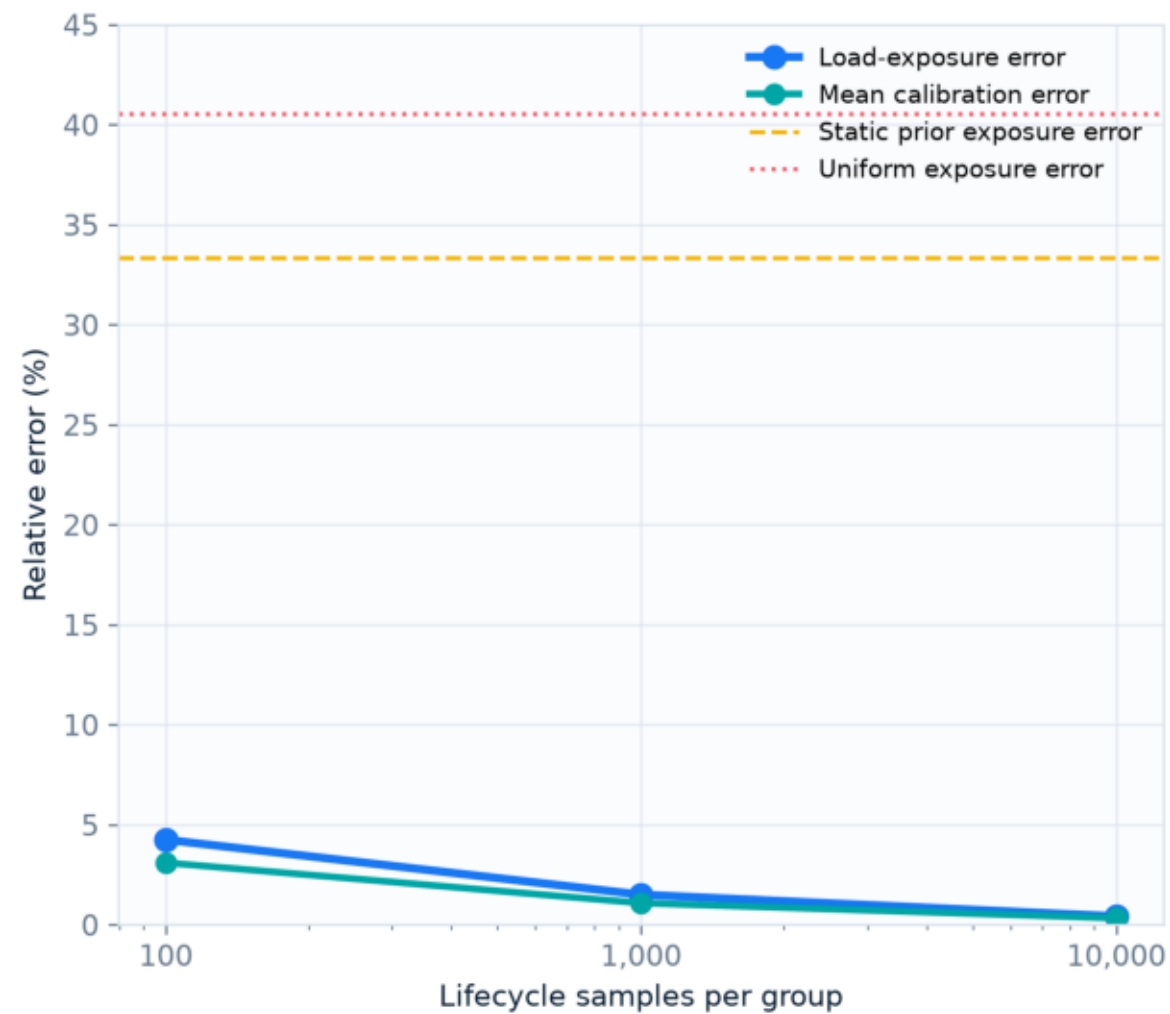


Interpretation: keep the regression visible, but do not present nine observations as a stable population estimate.

Seed 46003 generated + sealed - never evaluated or selected

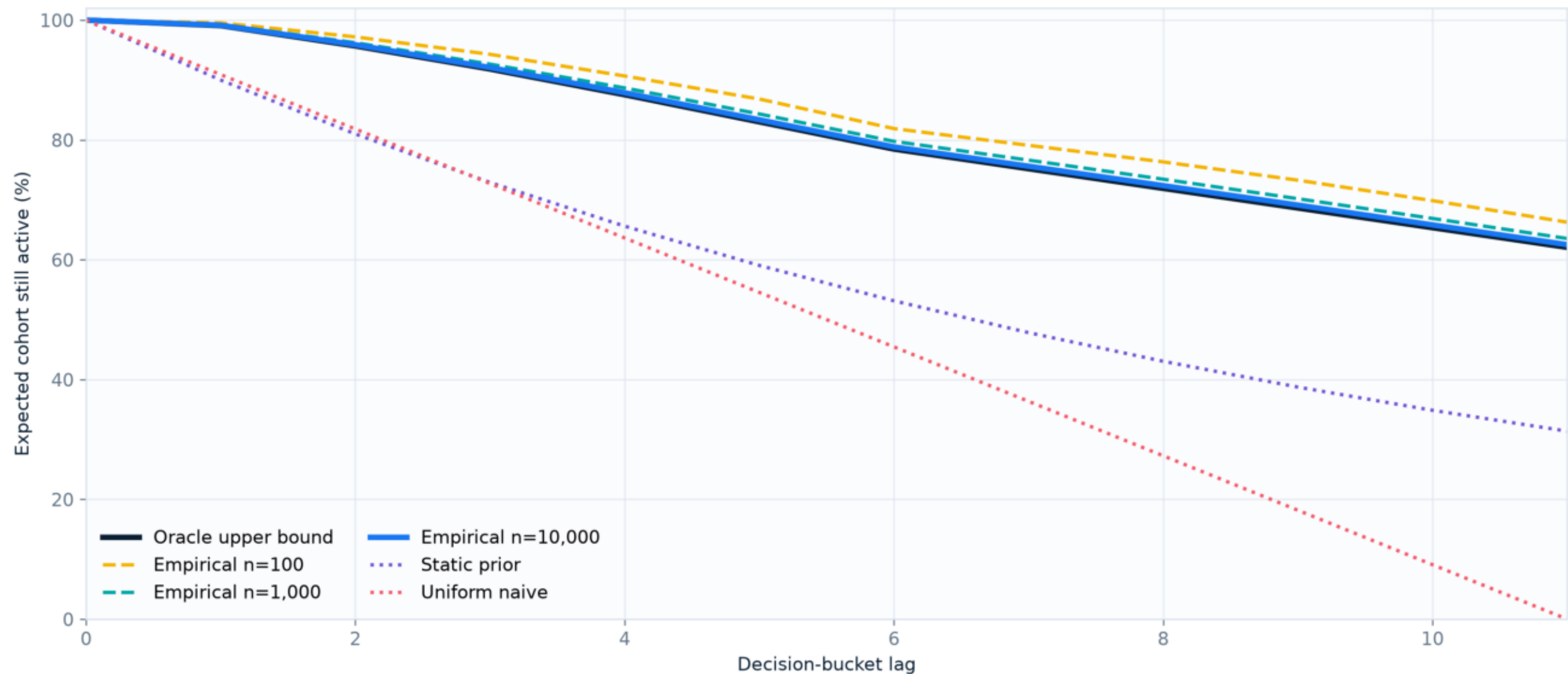
Kaplan-Meier mechanics converge on synthetic lifecycle telemetry

Synthetic censored records test estimation mechanics—not real-world distribution independence



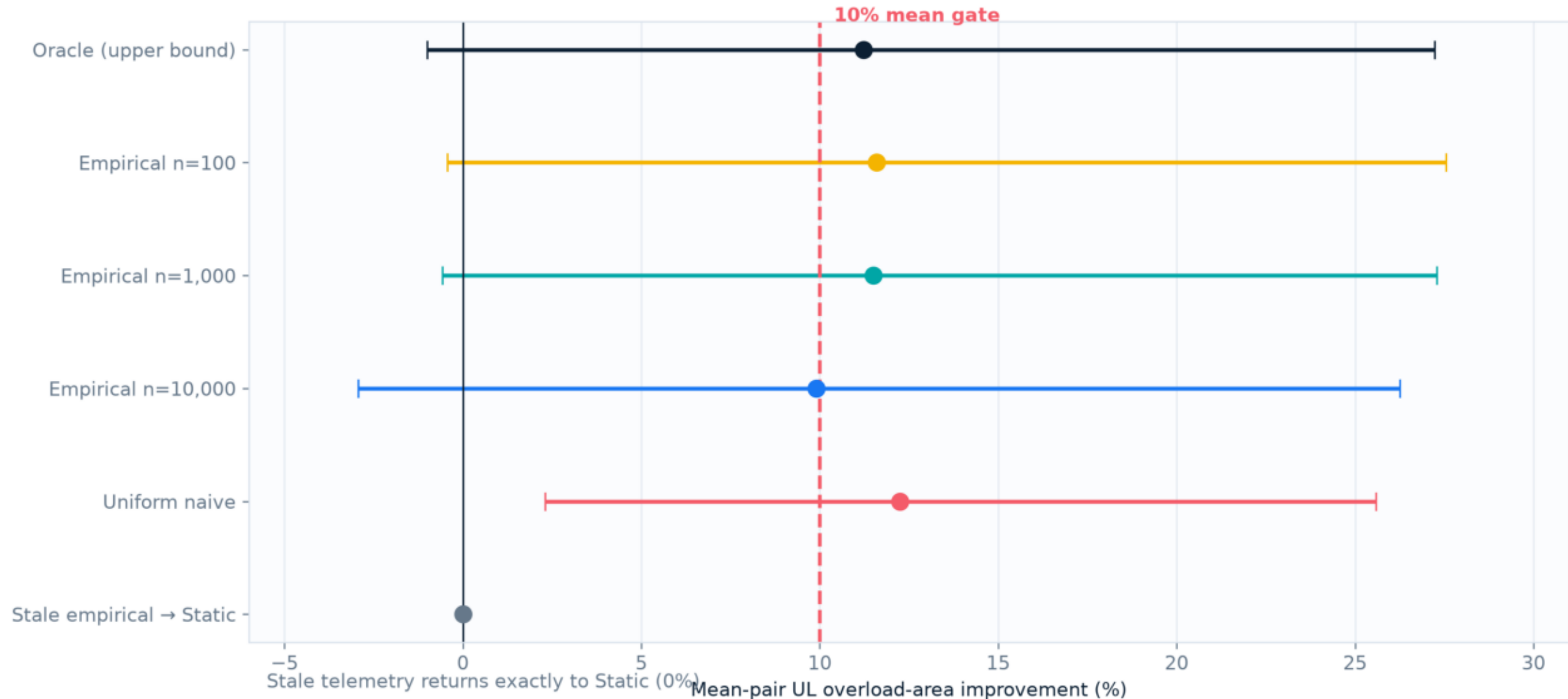
What the MPC actually consumes: expected cohort survival

Average across 96 traffic groups; empirical n=10,000 closely tracks the non-deployable oracle



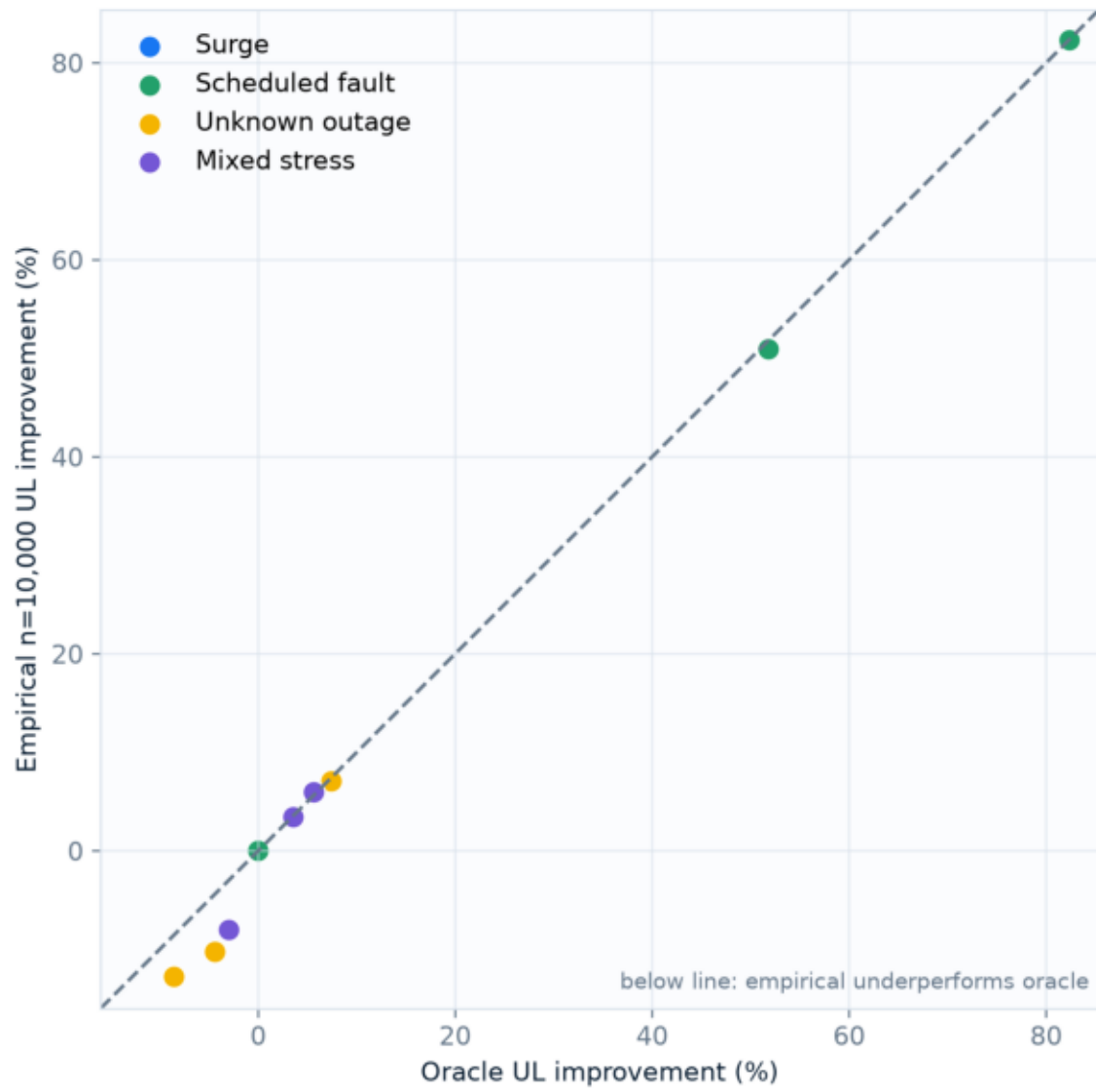
Empirical survival is relatively equivalent—not operationally robust

Oracle timeout 68.75% · empirical n=10,000 timeout 69.50% · same 12 development seeds



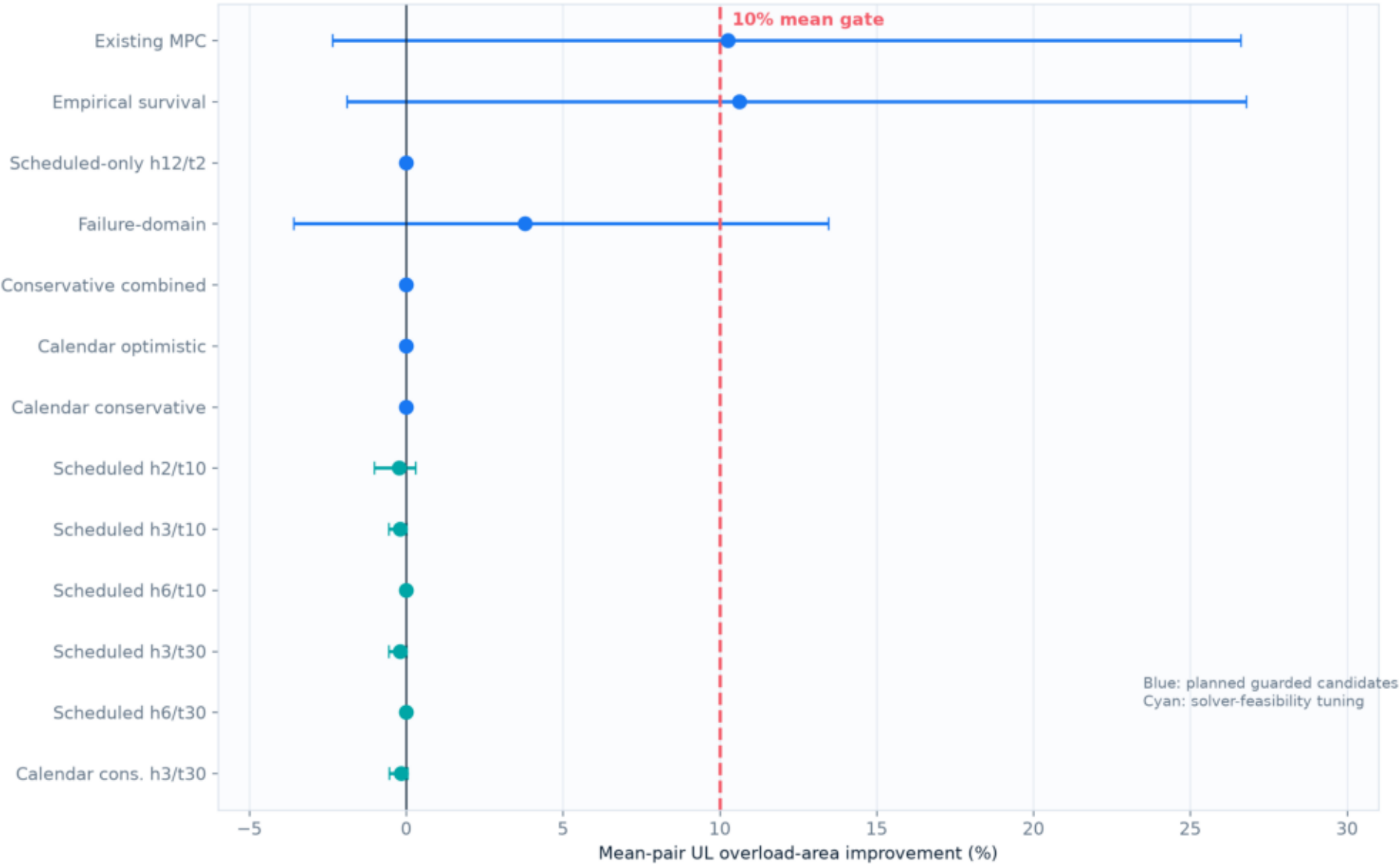
Imperfect survival follows oracle behavior pair by pair

Empirical n=10,000 versus oracle upper bound; 12 identical development pairs



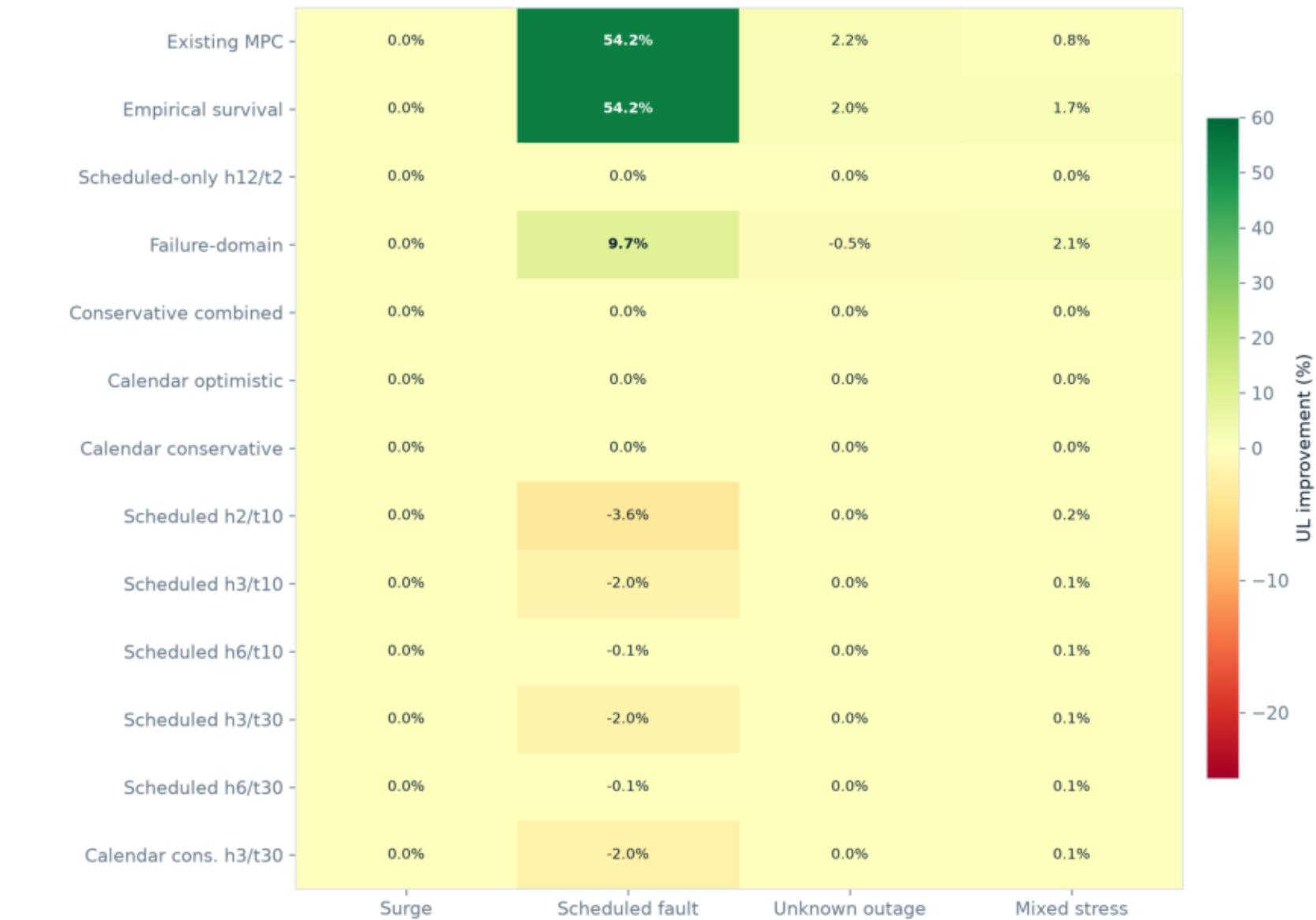
All 13 guarded MPC candidates fail the frozen Day-5 bar

Forest plot shows paired mean UL improvement and bootstrap 95% interval



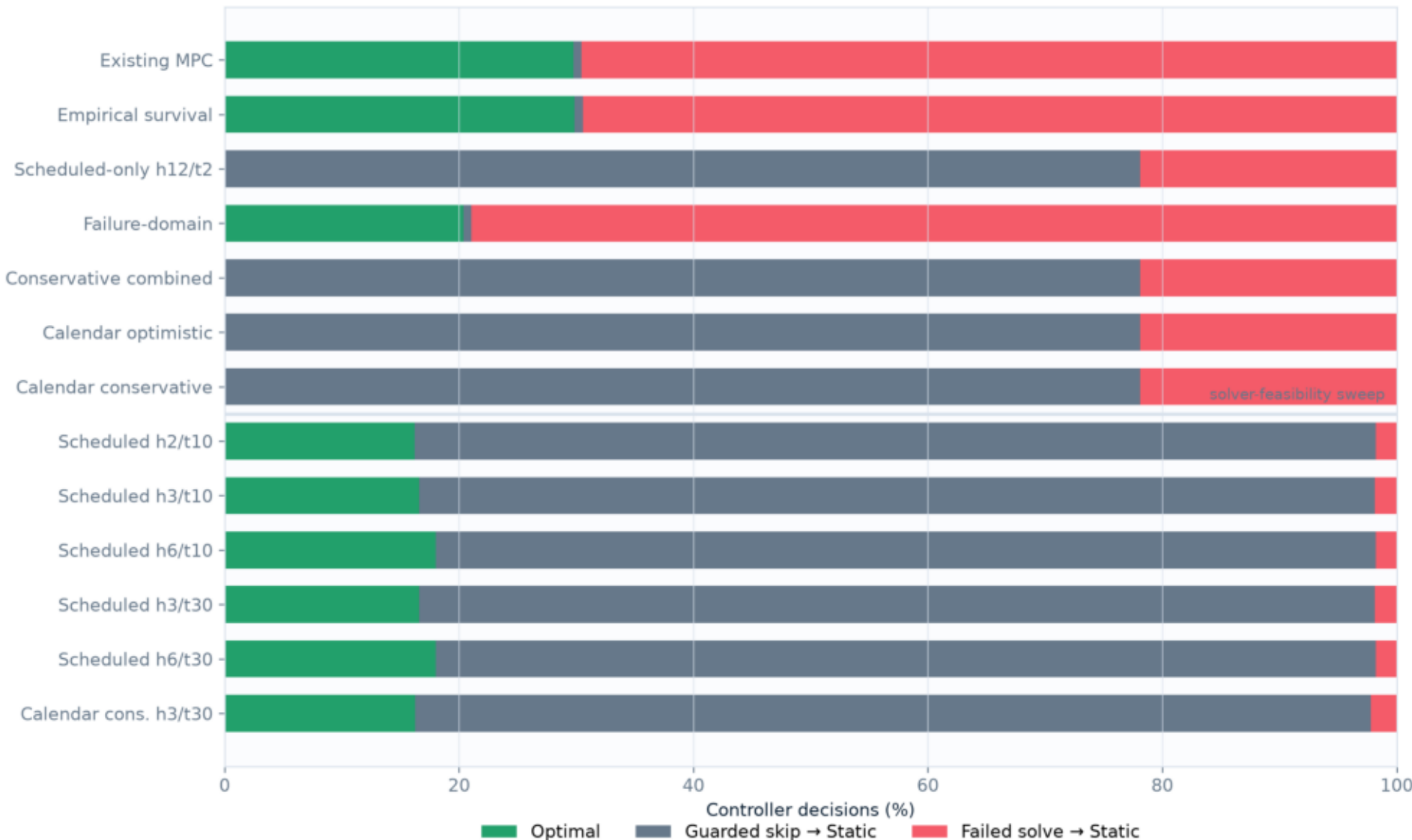
Scenario view: broad MPC gains are concentrated and unstable

Aggregate UL overload-area improvement; three paired seeds per scenario



Solver-feasibility tuning removed timeouts—not the performance gap

Share of controller decisions by solver status; failed = timeout, error, or infeasible



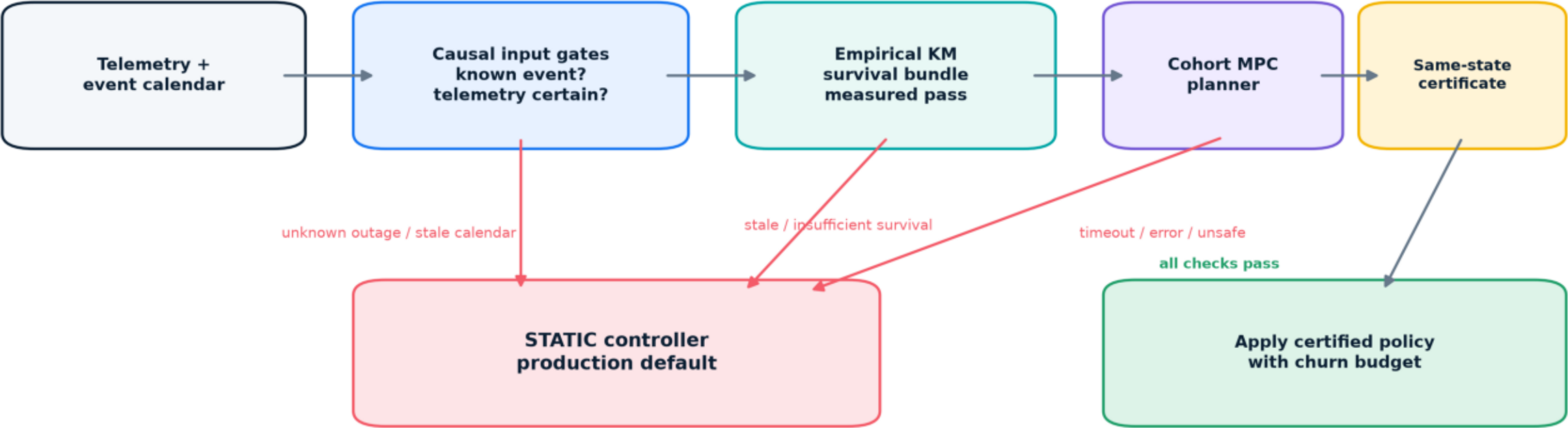
Fail closed means every gate must pass

Green = pass, red = fail; no candidate clears all 11 frozen development gates

Existing MPC	✓	✗	✓	✓	✗	✓	✗	✗	✓	✗	✓
Empirical survival	✓	✗	✓	✓	✗	✓	✗	✗	✓	✗	✓
Scheduled-only h12/t2	✗	✗	✗	✓	✓	✓	✗	✗	✓	✓	✓
Failure-domain	✗	✗	✓	✓	✗	✓	✗	✗	✓	✓	✓
Conservative combined	✗	✗	✗	✓	✓	✓	✗	✗	✓	✓	✓
Calendar optimistic	✗	✗	✗	✓	✓	✓	✗	✗	✓	✓	✓
Calendar conservative	✗	✗	✗	✓	✓	✓	✗	✗	✓	✓	✓
Scheduled h2/t10	✗	✗	✗	✓	✓	✗	✓	✗	✓	✓	✓
Scheduled h3/t10	✗	✗	✗	✓	✓	✗	✓	✗	✓	✓	✓
Scheduled h6/t10	✗	✗	✓	✓	✓	✓	✓	✗	✓	✓	✓
Scheduled h3/t30	✗	✗	✗	✓	✓	✗	✓	✗	✓	✓	✓
Scheduled h6/t30	✗	✗	✓	✓	✓	✓	✓	✗	✓	✓	✓
Calendar cons. h3/t30	✗	✗	✗	✓	✓	✗	✗	✗	✓	✓	✓
	Mean UL $\geq 10\%$	Bootstrap lower > 0	Severity-weighted > 0	Combined severity-weighted unknown + mixed $\geq -2\%$	Worst pair $> -10\%$	No DL/drop/session regression	No timeout/error	Unexpected fallback $\leq 1\%$	Skipped decisions $\leq 95\%$	Churn ≤ 0.05	Measured survival robust

Production architecture: MPC is an optional, guarded branch

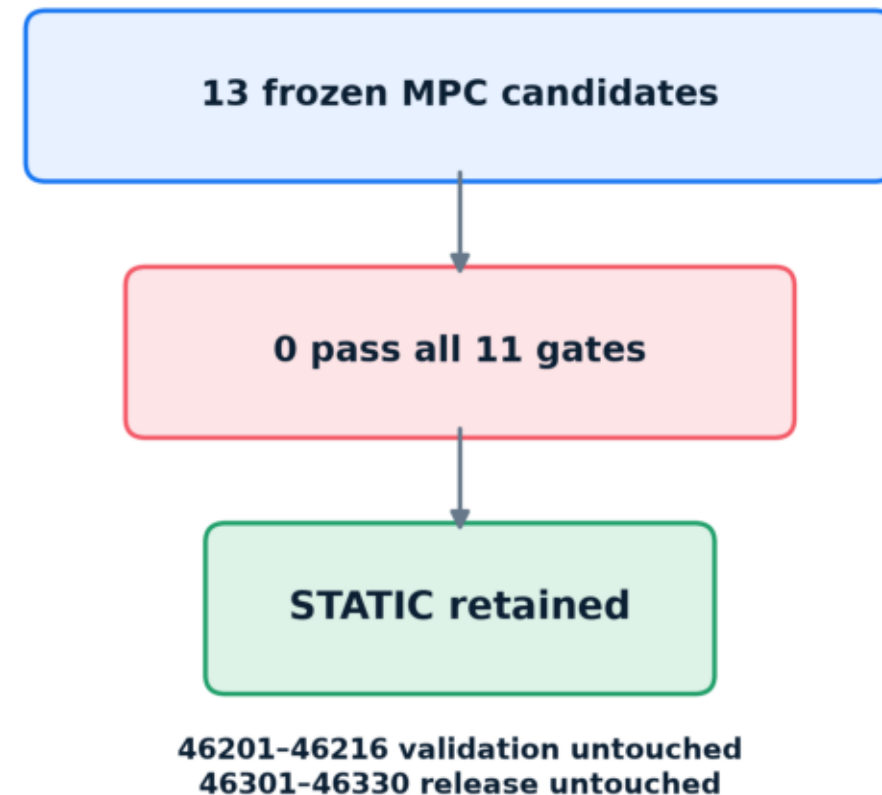
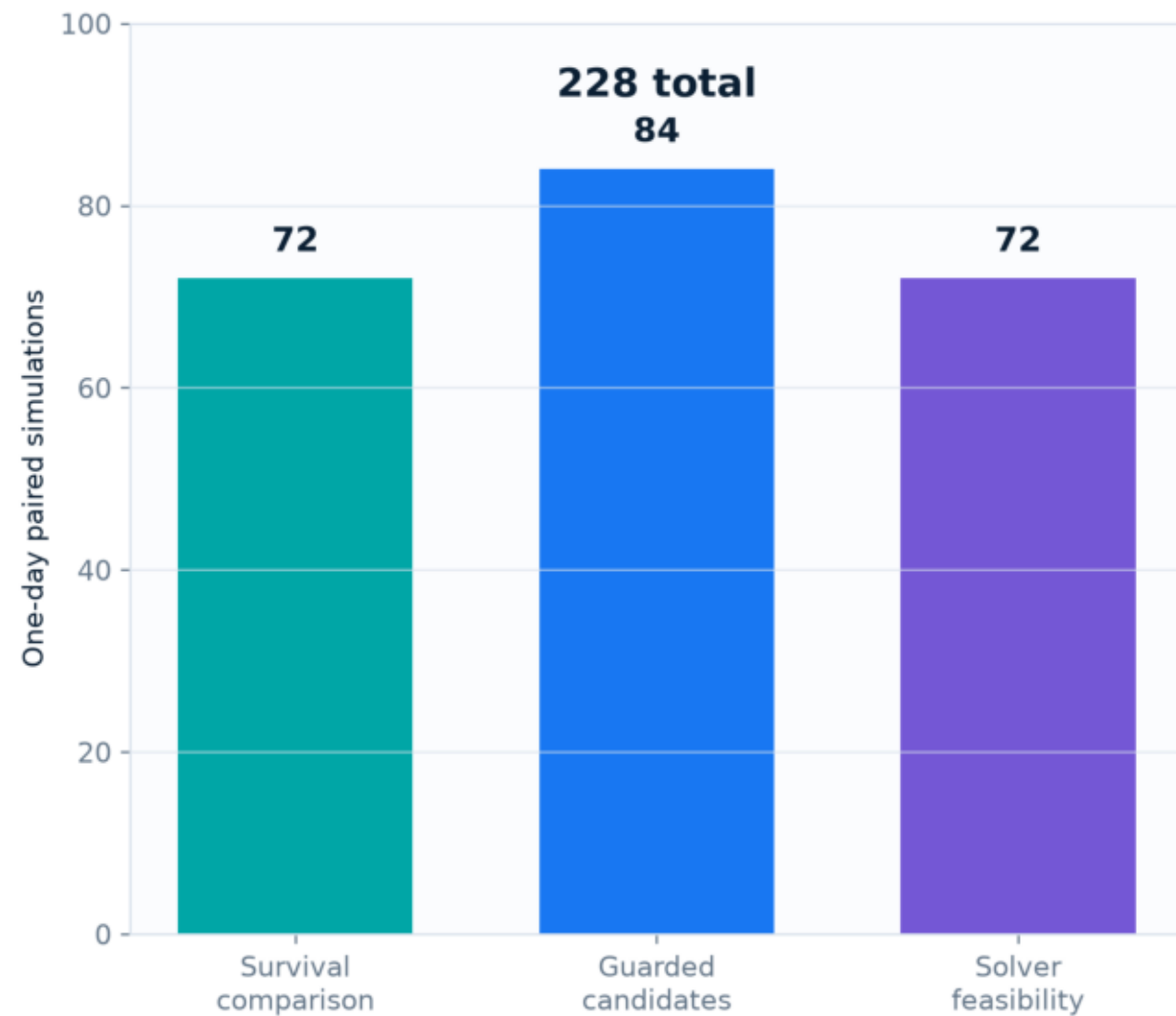
Unknown outages, uncertain telemetry, stale survival, or failed certification return immediately to Static



Measured result: the guardrail plumbing works; no tested MPC branch earned authority to replace Static.

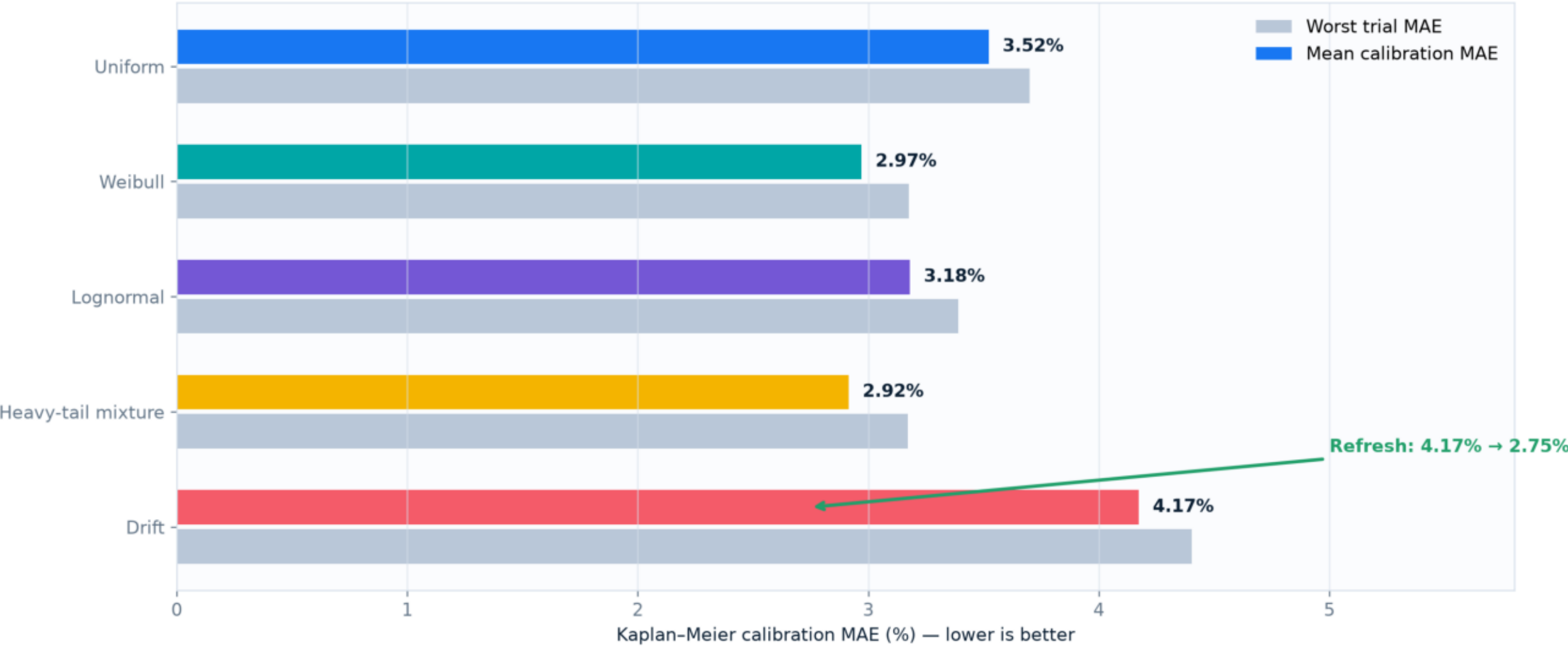
228 paired one-day runs support a disciplined stop decision

The campaign used development seeds repeatedly and preserved validation/release for a genuine winner



Distribution-blind lifecycle survival works mechanically

125 auditor-hidden trials · fit sees only start, end, censor, class and timestamps

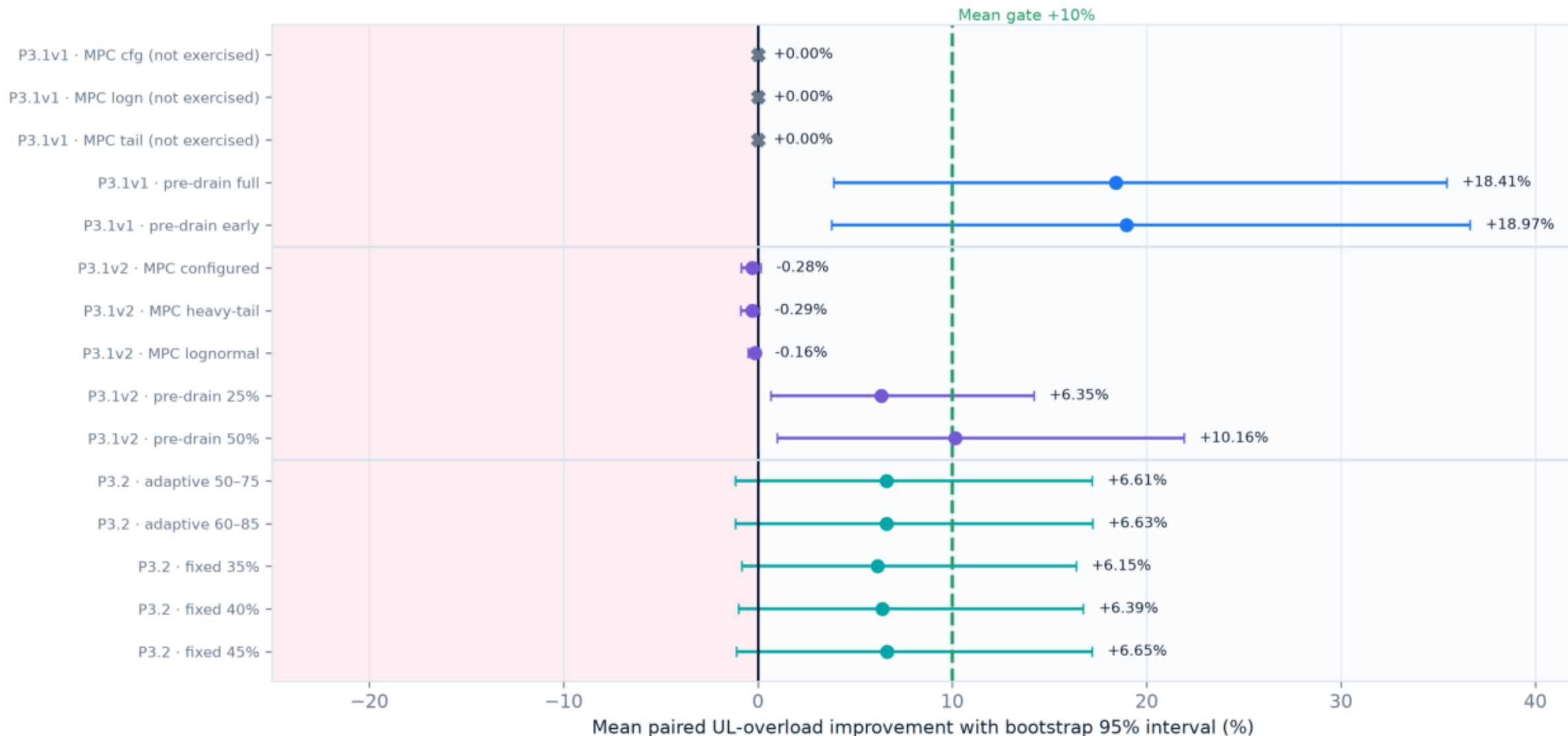


WORKED Staleness fail-closed 100% · pooling coverage 25%

BOUNDARY Simulated observations ≠ external real-world calibration

Every new development candidate: mean, uncertainty and mechanism status

360 paired one-day runs · confidence interval must exclude zero; mean must reach +10%

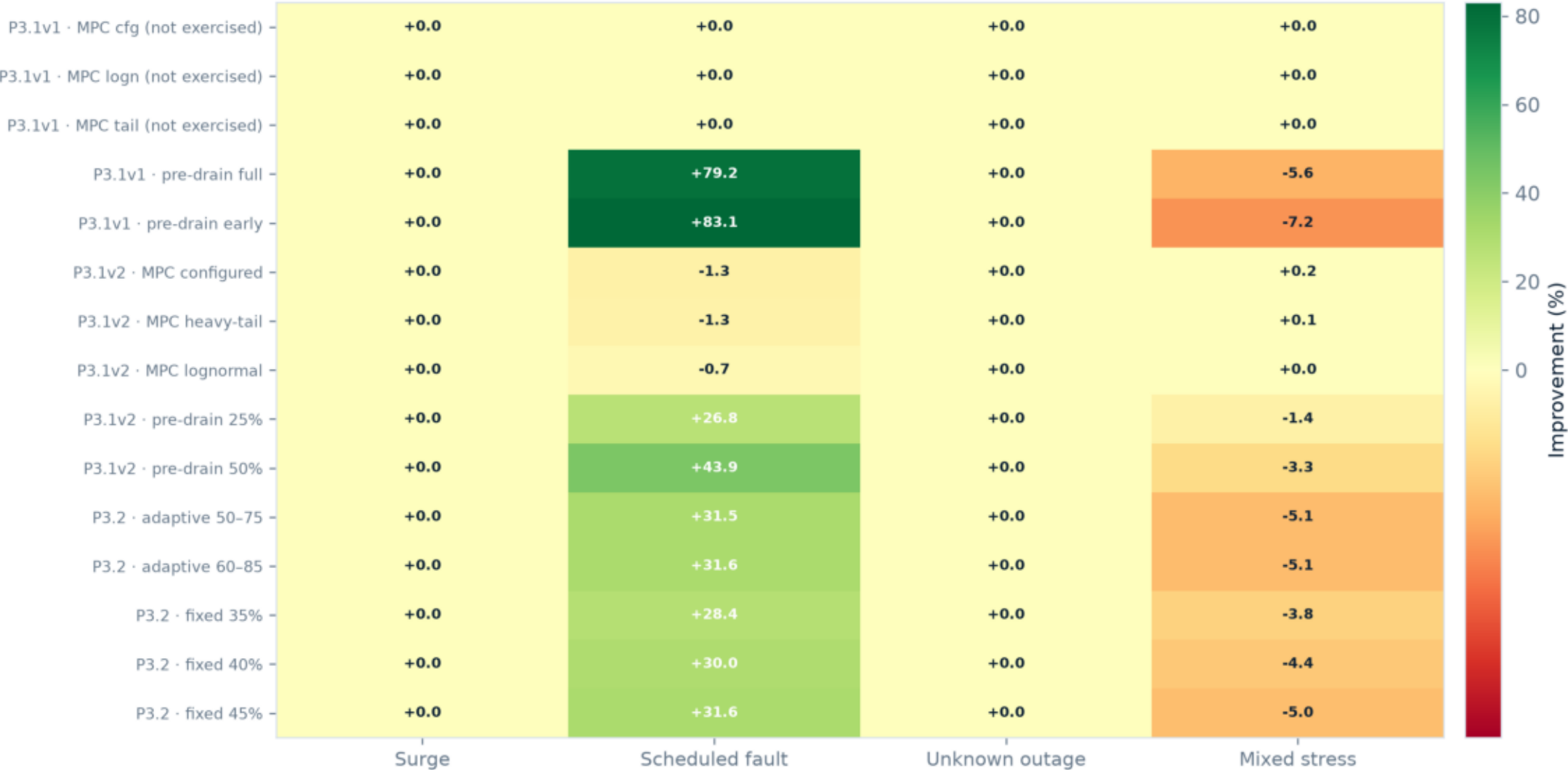


X = no proposed action: valid fail-closed, invalid mechanism exercise

Fresh development seeds 46401-46472 · validation/release untouched

Where gains came from—and where they failed

Scenario-stratified mean UL-overload improvement (%) · scheduled benefit did not generalize



WORKED: scheduled-fault pre-drain · DID NOT: robust transfer to mixed stress

Post-audit promotion scorecard: no all-green row

The added zero-overflow gate rejects every pre-drain family with predicted slack

P3.1v1 · MPC cfg (not exercised)	✗	✗	✗	✓	✓	✓	✓	✓	✗	✓	✗	—	✓
P3.1v1 · MPC logn (not exercised)	✗	✗	✗	✓	✓	✓	✓	✓	✗	✓	✗	—	✓
P3.1v1 · MPC tail (not exercised)	✗	✗	✗	✓	✓	✓	✓	✓	✗	✓	✗	—	✓
P3.1v1 · pre-drain full	✓	✓	✗	✓	✗	✓	✓	✓	✓	✗	✓	—	✓
P3.1v1 · pre-drain early	✓	✓	✗	✓	✗	✓	✓	✓	✓	✗	✓	—	✓
P3.1v2 · MPC configured	✗	✗	✓	✓	✓	✗	✓	✓	✓	✓	✗	—	✓
P3.1v2 · MPC heavy-tail	✗	✗	✓	✓	✓	✗	✓	✓	✓	✓	✗	—	✓
P3.1v2 · MPC lognormal	✗	✗	✓	✓	✓	✗	✓	✓	✓	✓	✗	—	✓
P3.1v2 · pre-drain 25%	✗	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	—	✗
P3.1v2 · pre-drain 50%	✓	✓	✓	✓	✗	✓	✓	✓	✓	✓	✓	—	✗
P3.2 · adaptive 50-75	✗	✗	✓	✓	✗	✓	✓	✓	✓	✓	✓	✗	✗
P3.2 · adaptive 60-85	✗	✗	✓	✓	✗	✓	✓	✓	✓	✓	✓	✗	✗
P3.2 · fixed 35%	✗	✗	✓	✓	✓	✓	✓	✓	✓	✓	✓	✗	✗
P3.2 · fixed 40%	✗	✗	✓	✓	✗	✓	✓	✓	✓	✓	✓	✓	✗
P3.2 · fixed 45%	✗	✗	✓	✓	✗	✓	✓	✓	✓	✓	✓	✗	✗
	Mean UL $\geq 10\%$	Bootstrap lower >0	Severity-weighted >0	Combined severity-weighted unknown + mixed $\geq -2\%$	Worst pair $> -10\%$	No DL/drop/session regression	No timeout/error	Unexpected fallback $\leq 1\%$	Skipped decisions $\leq 95\%$	Churn ≤ 0.05	Measured survival robust	End-to-end deadline	Zero predicted overflow

Pre-drain benefit-tail frontier did not produce a winner

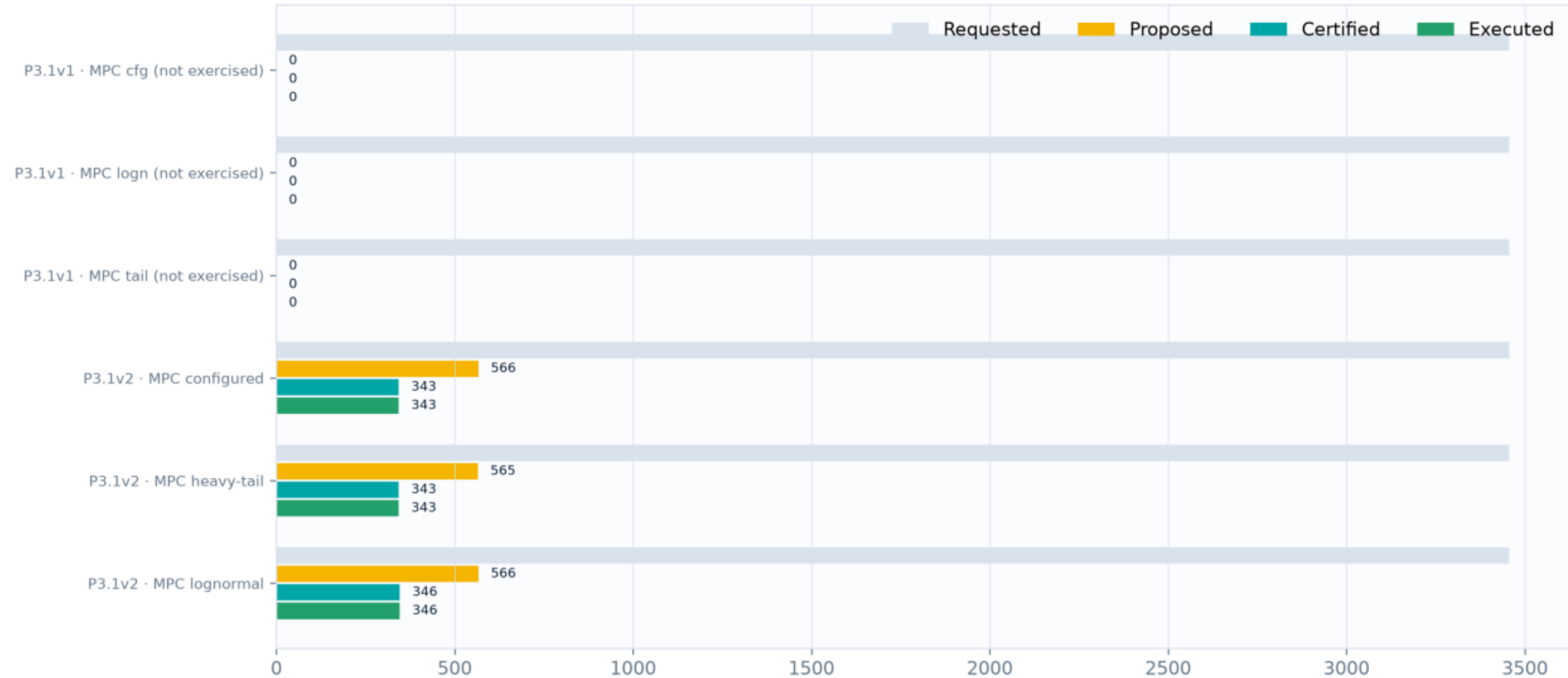
Point size = routing churn · promotion requires mean $\geq 10\%$ and worst pair $> -10\%$



v2 50%: enough mean, tail miss · v2 25% / P3.2: safer, insufficient/uncertain mean

MPC mechanism funnel: v1 was inert; v2 exercised and still did not help

Requested → proposed → certified → executed actions across 24 paired runs per candidate



Decision/action count

WORKED

Preflight enabled 1,697 proposals / 1,032 executions

DID NOT

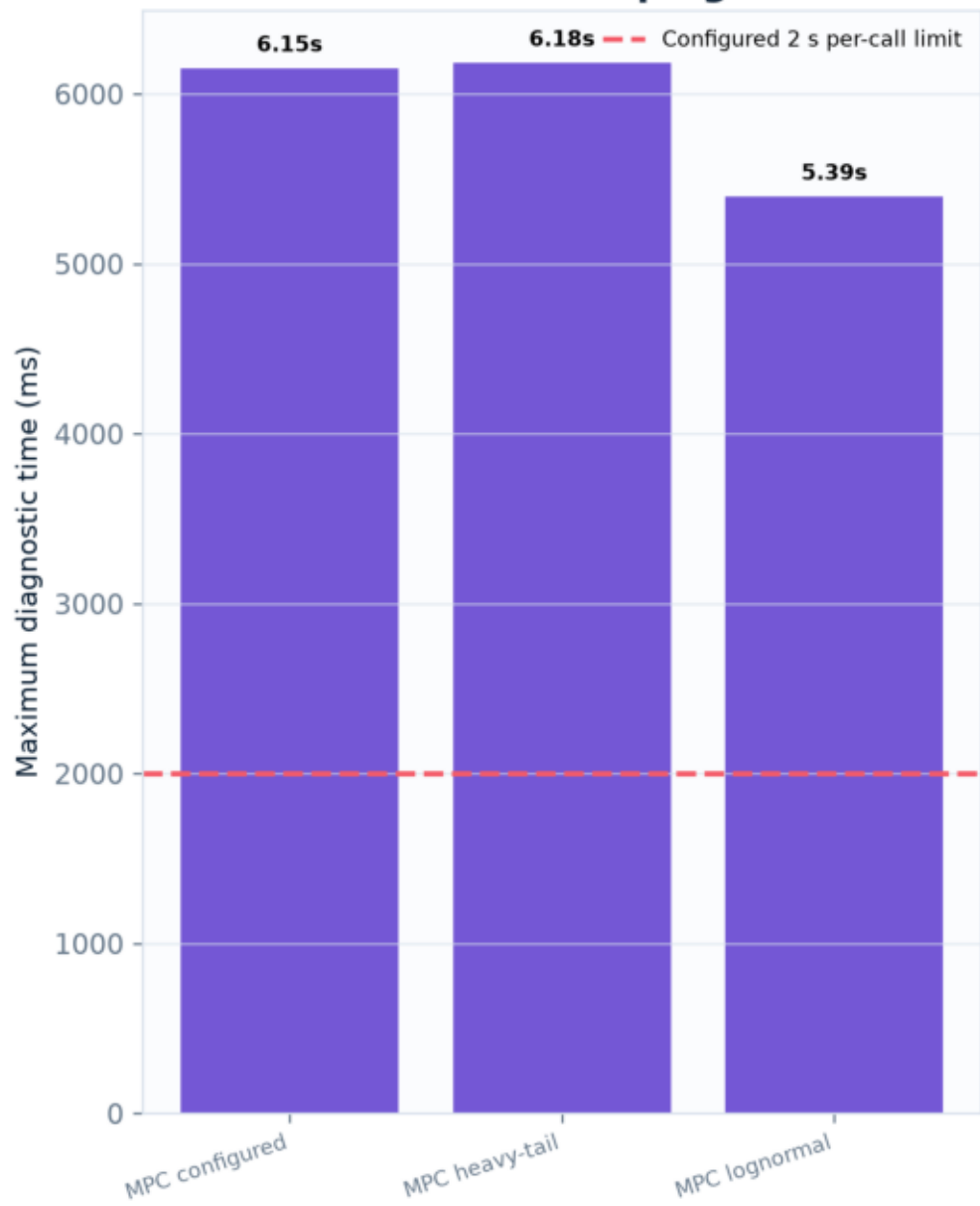
Exercised MPC averaged -0.16% to -0.29%

Fresh development seeds 46401-46472 · validation/release untouched

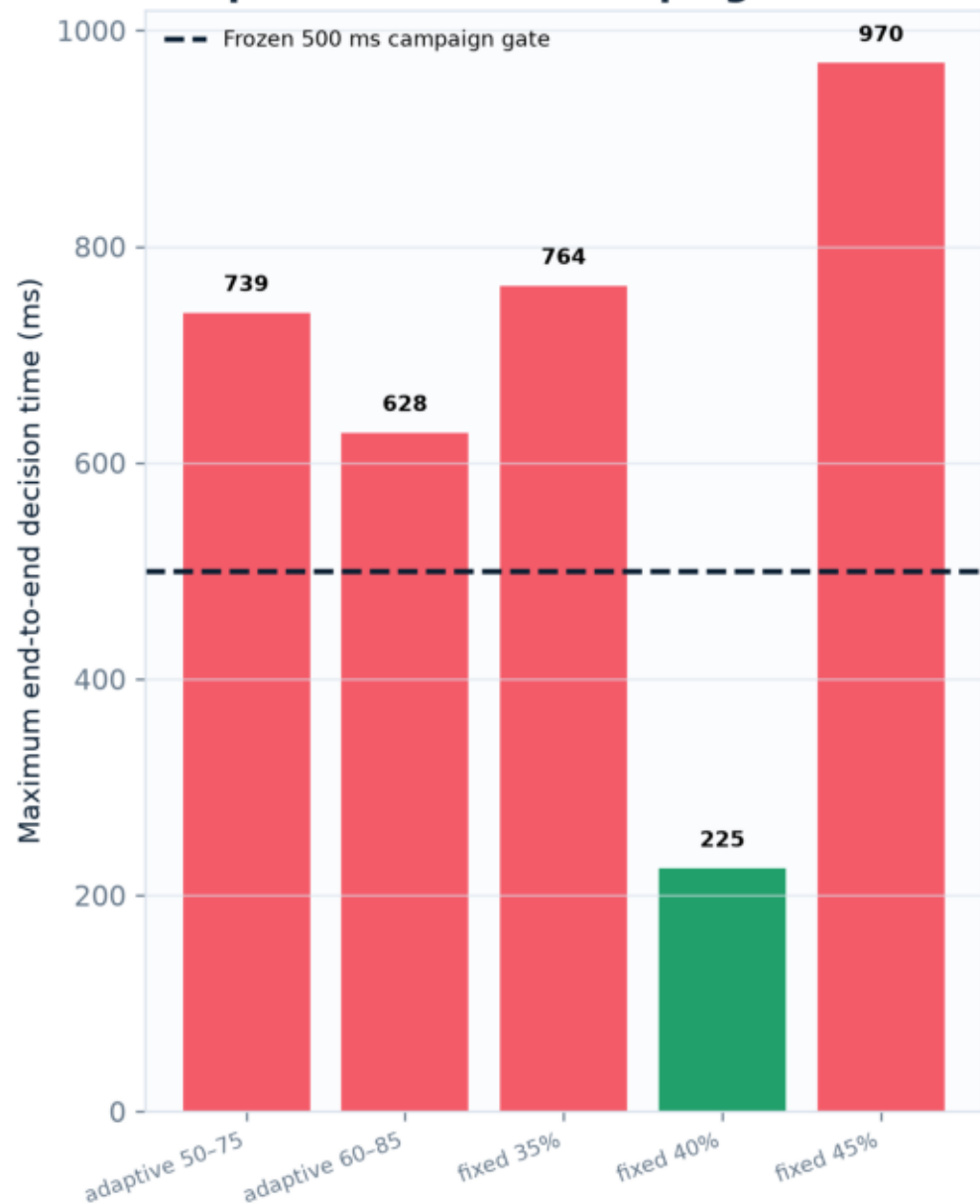
Campaign-saturation latency—not an isolated production benchmark

120 concurrent simulations on a saturated 125-CPU node; useful stress evidence only

P3.1 v2 MPC under campaign saturation



P3.2 pre-drain under campaign saturation

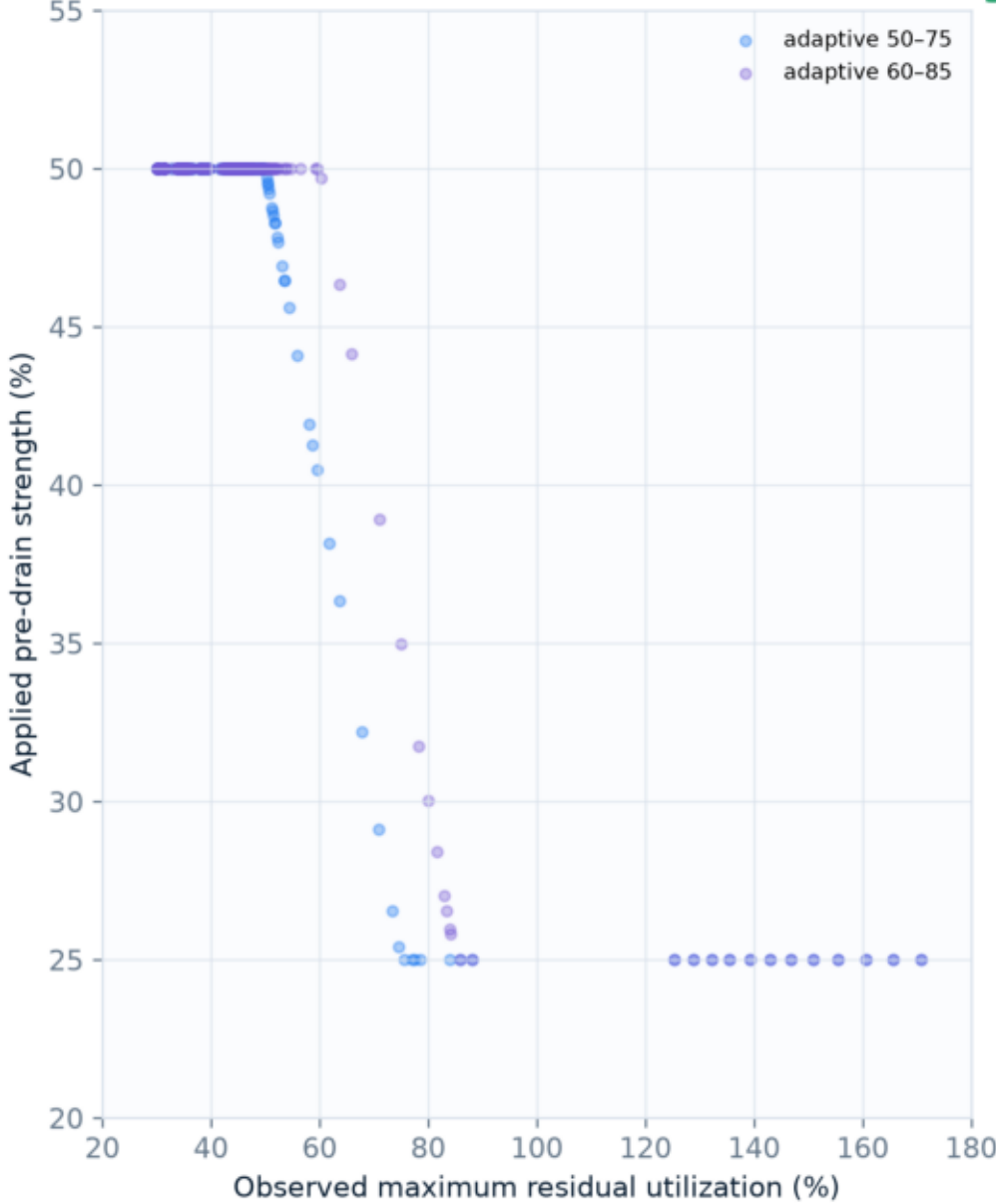


Fresh development seeds 46401-46472 · validation/release untouched

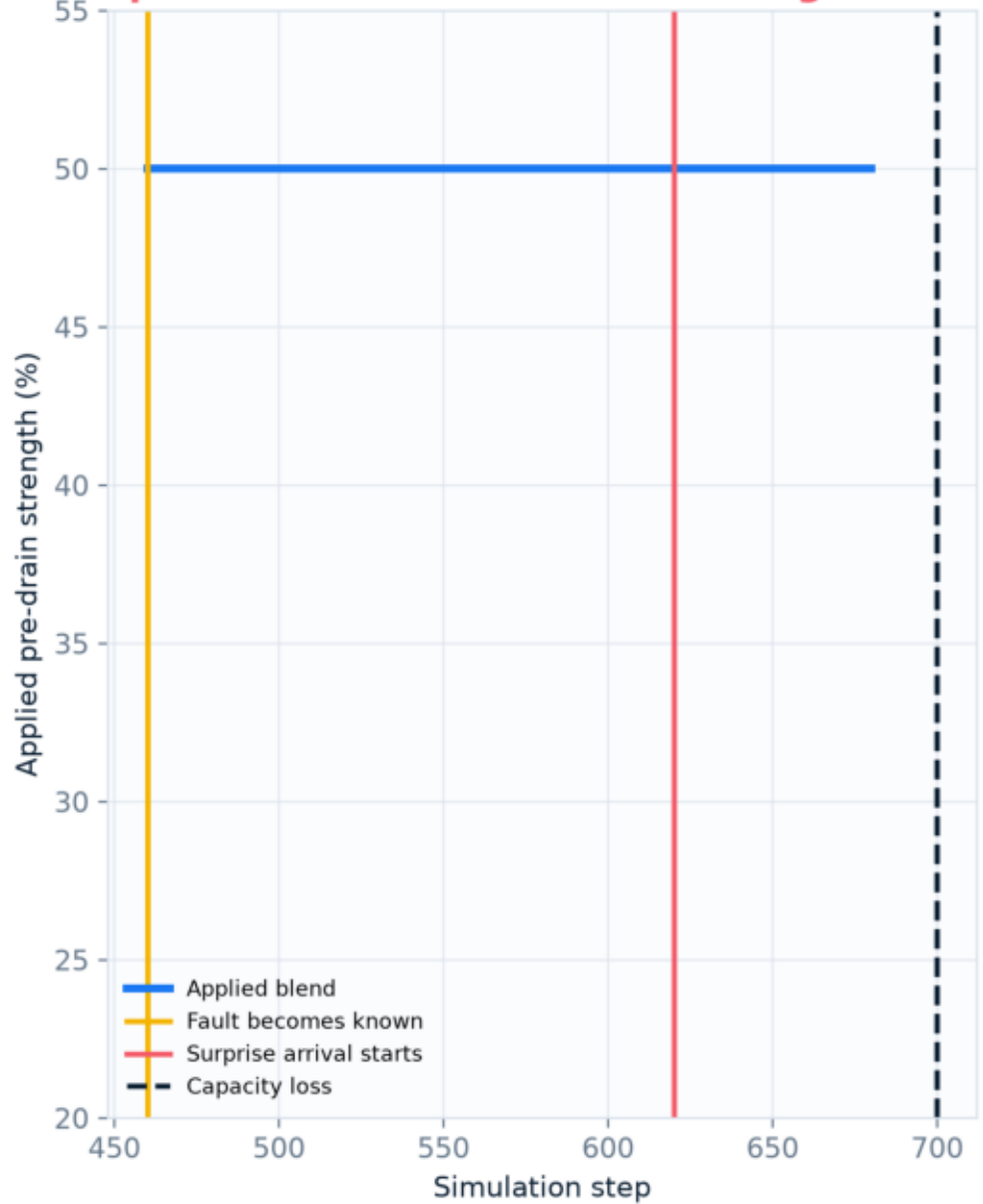
Adaptive pre-drain worked as coded—but causal lag defeated the tail hypothesis

Applied blend responds to observed residual utilization; worst pair's surprise begins after commitment starts

Mechanism exercised across full 25-50% range



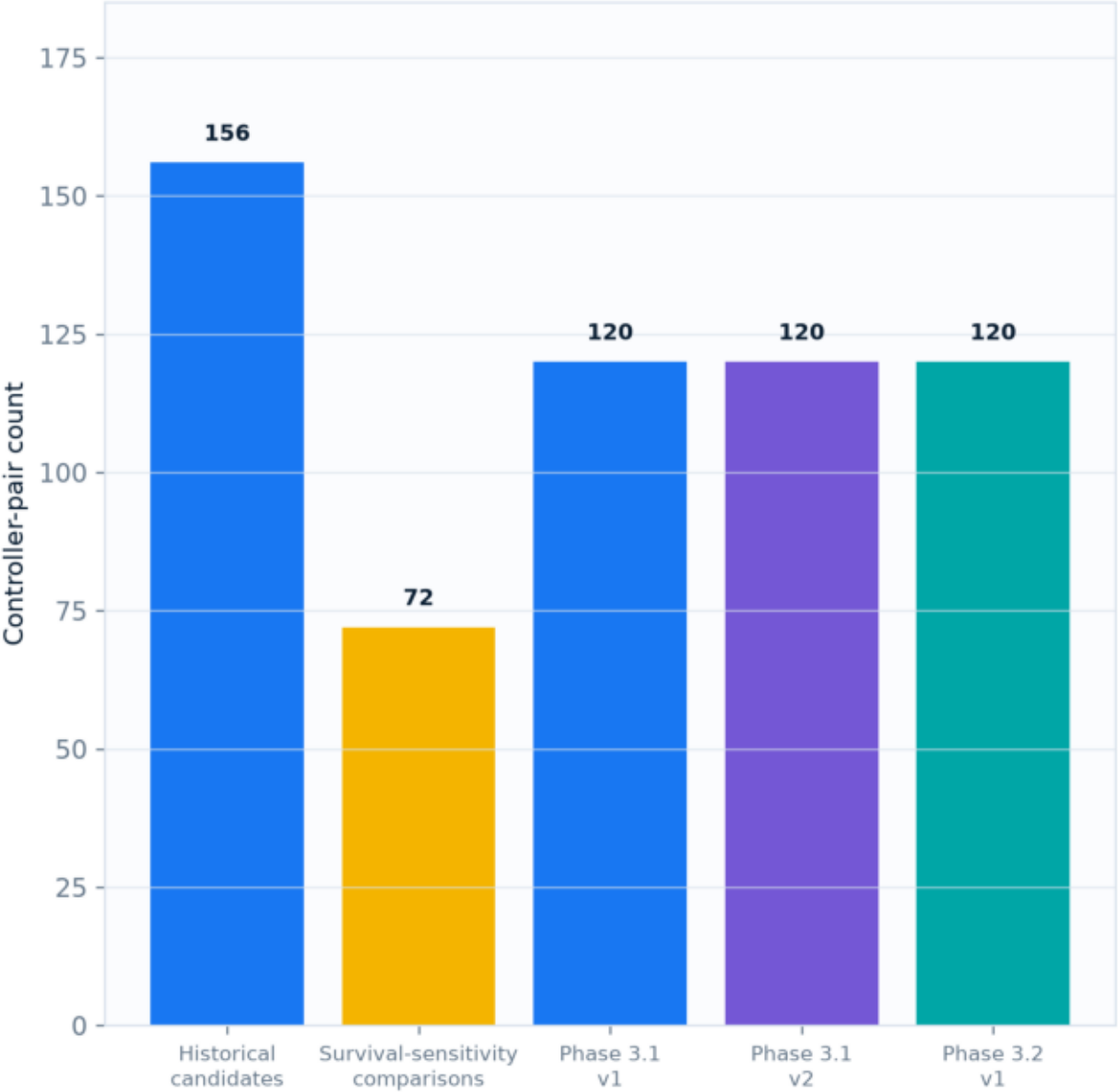
Worst pair seed 46468: full 50% throughout lead-up



Exact controller-pair inventory and seed firewall

516 declared-candidate pairs + 72 survival-sensitivity comparisons = 588 total

No ambiguity: all 588 pairs classified



Seed firewall held

Development consumed as registered
46101-46112 · 46401-46472

Validation UNTOUCHED
46201-46216

Release UNTOUCHED
46301-46330

Forecast test generated, never evaluated
46003

What worked—and what did not

Post-audit corrections change labels, not outcomes

WORKED



Reproducibility

Historical freezes retained; v6 corrections separately hashed and archived



Lifecycle estimator

Observable export + Kaplan-Meier calibrated across 5 hidden distributions



Overflow safety

Nonzero predicted slack now records by resource and fails closed to Static



Oracle fixture

Authoritative completed-job evaluation restored; full suite is green



Scheduled headroom

Pre-drain repeatedly reduced scheduled-fault overload in replay



Release discipline

516 candidate pairs + 72 sensitivity pairs; zero unsafe promotions

DID NOT WORK



Forecast promotion

Useful gains remained below the frozen 15% target



MPC benefit

Exercised variants were neutral/slightly harmful



Generalization

Scheduled benefit did not transfer safely to mixed stress



Tail safety

Strong blends missed the –10% worst-pair gate



Historical certification

Optimal-with-overflow actions are reclassified as non-certified



Latency claim

225-970 ms reflects saturated campaign execution, not production isolation

FINAL DECISION · RETAIN STATIC · SHADOW/REPLAY ONLY FOR MPC/PRE-DRAIN

Fresh development seeds 46401-46472 · validation/release untouched

Final decision

Retain Static

MPC and pre-drain: guarded shadow/replay only.

Predicted overflow: fail closed, report slack, never certify.

No protected validation or release seed was consumed.

No post-hoc blend interpolation was run.

Next: C-DOT demo integration · Static control · 12-node/384-shard rehearsal